5 Induction and Recursion

- **5.1 Mathematical Induction**
- **5.2 Strong Induction and Well-Ordering**
- **5.3 Recursive Definitions and Structural Induction**
- 5.4 Recursive Algorithms
- 5.5 Program Correctness

Inductive Proofs

Rosen 8th ed., § 5.1

§ 5.1: Mathematical Induction

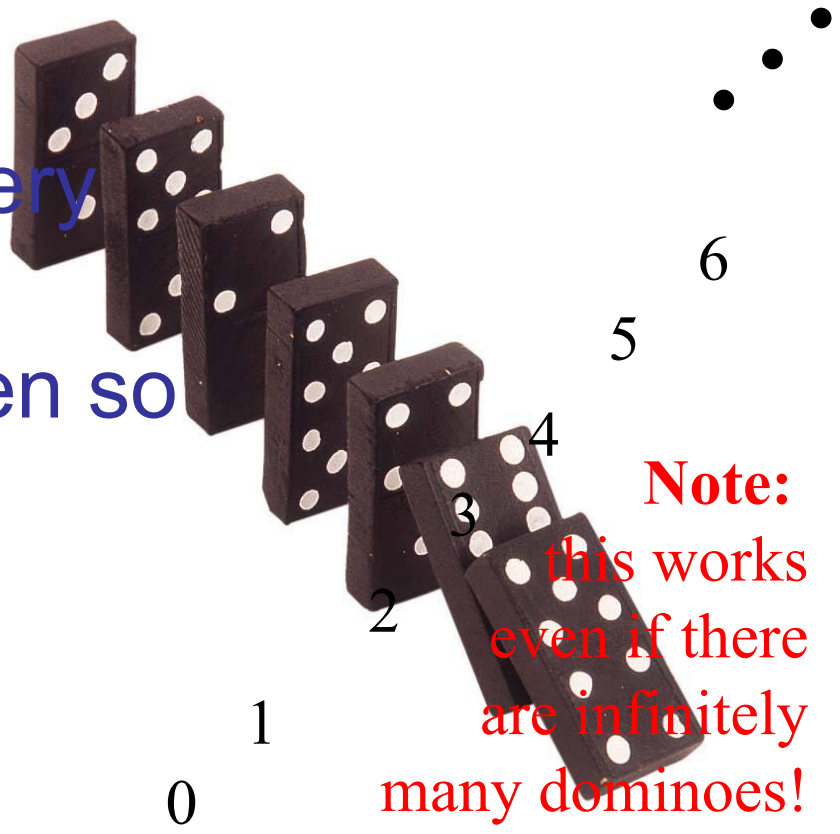
- A powerful, rigorous technique for proving that a predicate $P(n)$ is true for *every* natural number n, no matter how large.
- Essentially a "domino effect" principle.
- Based on a predicate-logic inference rule:

$$\frac{P(0) \qquad \forall n \geq 0\ (P(n) \rightarrow P(n+1))}{\therefore \forall n \geq 0\ P(n)}$$

"The First Principle of Mathematical Induction"

The "Domino Effect"

- **Premise #1:** Domino #0 falls.
- **Premise #2:** For every $n \in \mathbf{N}$, if domino #n falls, then so does domino #n+1.
- **Conclusion:** All of the dominoes fall down!



Note: this works even if there are infinitely many dominoes!

Validity of Induction

Proof that $\forall k{\geq}0\ P(k)$ is a valid consequent:

Given any $k{\geq}0$, the 2nd antecedent $\forall n{\geq}0\ (P(n){\rightarrow}P(n{+}1))$ trivially implies that $\forall n{\geq}0\ (n{<}k){\rightarrow}(P(n){\rightarrow}P(n{+}1))$, *i.e.*, that $(P(0){\rightarrow}P(1)) \wedge (P(1){\rightarrow}P(2)) \wedge \ldots \wedge (P(k{-}1){\rightarrow}P(k))$. Repeatedly applying the hypothetical syllogism rule to adjacent implications in this list $k{-}1$ times then gives us $P(0){\rightarrow}P(k)$; which together with $P(0)$ (antecedent #1) and *modus ponens* gives us $P(k)$. Thus $\forall k{\geq}0\ P(k)$. ■

Outline of an Inductive Proof

- Let us say we want to prove $\forall n\ P(n)$…
  - Do the *base case* (or *basis step*): Prove $P(0)$.
  - Do the *inductive step*: Prove $\forall n\ P(n) \rightarrow P(n+1)$.
    - *E.g.* you could use a direct proof, as follows:
    - Let $n \in \mathbf{N}$, assume $P(n)$. (*inductive hypothesis*)
    - Now, under this assumption, prove $P(n+1)$.
  - The inductive inference rule then gives us $\forall n\ P(n)$.

Generalizing Induction

- Rule can also be used to prove $\forall n \geq c\ P(n)$ for a given constant $c \in \mathbf{Z}$, where maybe $c \neq 0$.
  - In this circumstance, the base case is to prove $P(c)$ rather than $P(0)$, and the inductive step is to prove $\forall n \geq c\ (P(n) \rightarrow P(n+1))$.
- Induction can also be used to prove $\forall n \geq c\ P(a_n)$ for any arbitrary series $\{a_n\}$.
- Can reduce these to the form already shown.

Induction Example (1st princ.)

- Prove that the sum of the first *n* odd positive integers is n^2. That is, prove:

$$\forall n \geq 1 : \underbrace{\sum_{i=1}^{n} (2i-1) = n^2}_{P(n)}$$

- Proof by induction.
  - Base case: Let *n*=1. The sum of the first 1 odd positive integer is 1 which equals 1^2. (Cont…)

Example cont.

- Inductive step: Prove $\forall n \geq 1$: $P(n) \rightarrow P(n+1)$.
  - Let $n \geq 1$, assume $P(n)$, and prove $P(n+1)$.

$$\sum_{i=1}^{n+1}(2i-1) = \left(\sum_{i=1}^{n}(2i-1)\right) + (2(n+1)-1)$$

$$= n^2 + 2n + 1$$

$$= (n+1)^2$$

By inductive hypothesis P(n)

Another Induction Example

- Prove that $\forall n>0$, $n<2^n$. Let $P(n)=(n<2^n)$
  - Base case: $P(1)=(1<2^1)=(1<2)=\mathbf{T}$.
  - Inductive step: For $n>0$, prove $P(n)\rightarrow P(n+1)$.
    - Assuming $n<2^n$, prove $n+1 < 2^{n+1}$.
    - Note $n + 1 < 2^n + 1$ (by inductive hypothesis)
      $< 2^n + 2^n$ (because $1<2=2\cdot 2^0 \leq 2\cdot 2^{n-1}= 2^n$)
      $= 2^{n+1}$
    - So $n + 1 < 2^{n+1}$, and we're done.

Proof by Induction

- **Basis Step(** 基始）**.** Prove that *S*(k) is true for some (small) value of k.
- **Inductive Step**（归纳）**.** For every *n*, *assume* that *S*(*n*) is true and *prove* that *S*(*n*+ 1) is true.

- Together, these prove $\forall i \geq k : P(i)$

Fibonacci Numbers（斐波那契数列）

- The Fibonacci sequence 0, 1, 1, 2, 3, 5, ... is defined by the following formula

  F(0) = 0

  F(1) = 1

  F(n) = F(n-1) + F(n-2)

Fibonacci Sums

$0+1 = 1$

$0+1+1 = 2$

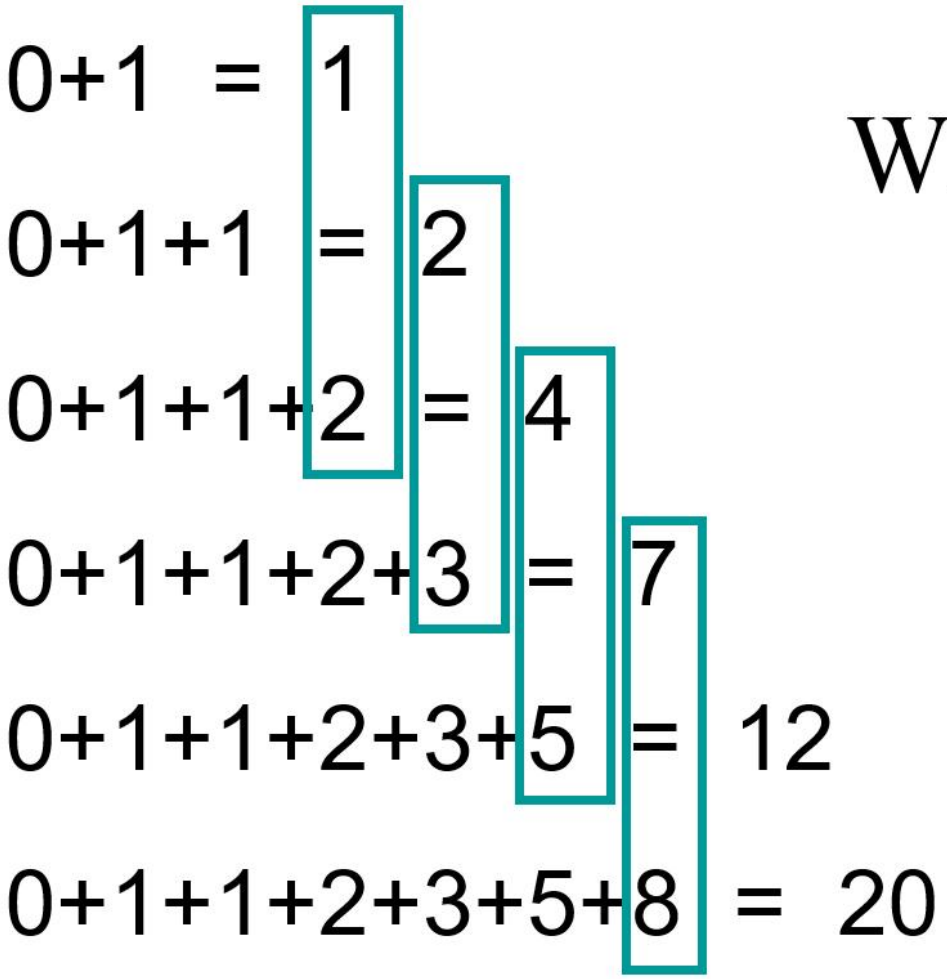
$0+1+1+2 = 4$

$0+1+1+2+3 = 7$

$0+1+1+2+3+5 = 12$

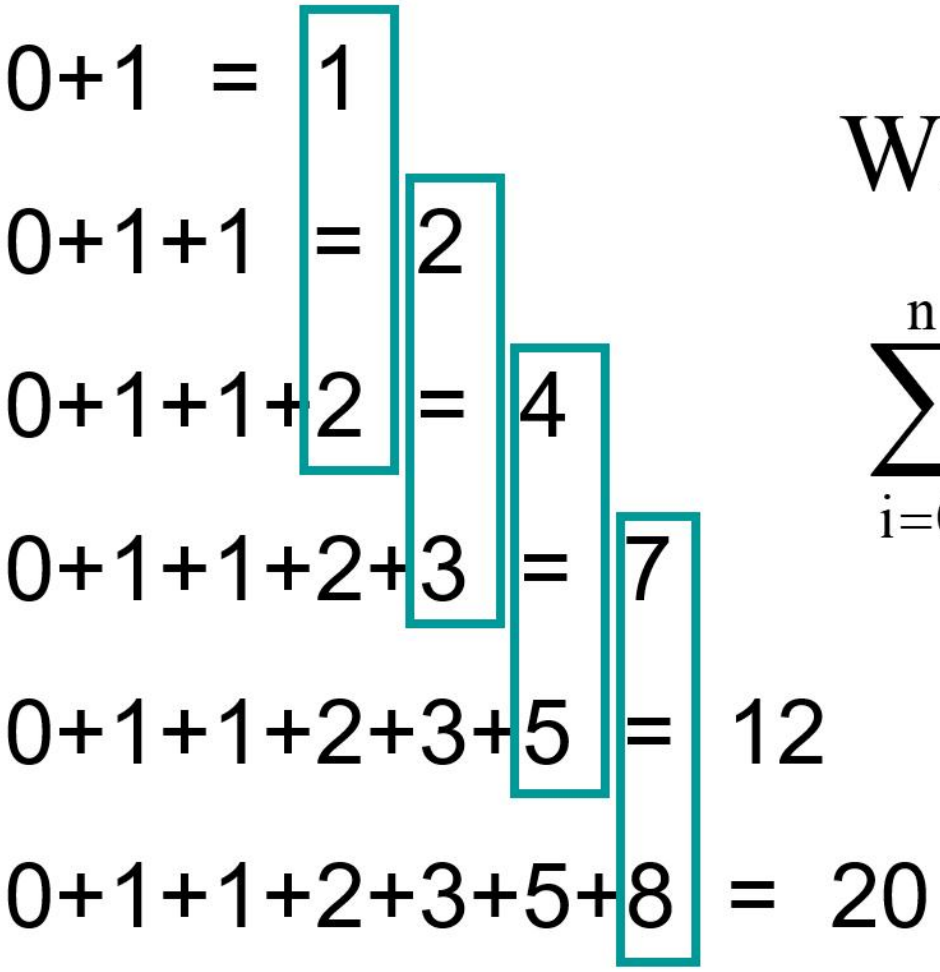
$0+1+1+2+3+5+8 = 20$

Fibonacci Sums



What's the pattern?

Fibonacci Sums



What's the pattern?

$$\sum_{i=0}^{n} F(i) = F(n+2) - 1$$

Fibonacci Sums

- Let's verify （证明）this using induction:
  - We need to verify a base case
  - Then we assume the formula （公式）holds for n and show it holds for n+1.

Base Case

N=0

$$\sum_{i=0}^{n} F(i) = 0$$

$$F(n+2) - 1 = 1 - 1$$

Inductive Step

$$\sum_{i=0}^{n+1} F(i)$$

$$= F(n+1) + \sum_{i=0}^{n} F(i)$$

$$= F(n+1) + F(n+2) - 1$$

$$= F(n+3) - 1$$

Assumption（假定）:

$$\sum_{i=0}^{n} F(i) = F(n+2) - 1$$

Conclusion（结论）

$$\sum_{i=0}^{n+1} F(i) = F(n+3) - 1$$

Example 1

- Show,by mathematical induction,that for all n≥1,
- Solution $1+2+3+\cdots+n=\frac{n(n+1)}{2}$
  - Let $p(n)$ be the predicate $1+2+3+\cdots+n=\frac{n(n+1)}{2}$
  - BS: n_0=1 , $1=\frac{1(1+1)}{2}$ $p(1)$ is true.
  - IS:assume for some fixed $k\geq 1$, $p(k)$ is true.

$$1+2+3+\cdots+(k+1)$$ the left-hand side of P(k+1)

$$=\frac{k(k+1)}{2}+(k+1)$$ using(1) to replace 1+2+$\cdots$ +k

$$=(k+1)[\frac{k}{2}+1]$$ factoring

$$=\frac{(k+1)(k+2)}{2}$$

$$=\frac{(k+1)((k+1)+1)}{2}$$ the right-hand side of P(k+1)

- *p*(*k*+*1*) is true.
- By the principle of mathematical induction,*p*(*n*) is true for $n \geq 1$.
- Q.E.D

Example 2

- Let $A_1, A_2 \ \ldots, \ A_n$ be any n sets. Show that

$$\overline{\left(\bigcup_{i=1}^{n} A_i \right)} = \bigcap_{i=1}^{n} \overline{A_i}$$

- Solution
  - Let $P(n)$ be the predicate that the equality hold for any n sets.

- BS: , $P(1)$ is true.
- IS: $\overline{A_1} = \overline{A_1}$

$$\overline{\left(\bigcup_{i=1}^{k+1} A_i\right)} = \overline{A_1 \cup A_2 \cup \cdots \cup A_k \cup A_{k+1}} \quad \text{left-hand side of P(k+1)}$$

$$= \overline{(A_1 \cup A_2 \cup \cdots \cup A_k) \cup A_{k+1}} \quad \text{associative property of } \cup$$

$$= \overline{(A_1 \cup A_2 \cup \cdots \cup A_k)} \cap \overline{A_{k+1}} \quad \text{by the De Morgan's law}$$

$$= \left(\bigcap_{i=1}^{k} \overline{A_i}\right) \cap \overline{A_{k+1}} \quad \text{using P(k)}$$

$$= \bigcap_{i=1}^{k+1} \overline{A_i} \quad \text{right-hand side of P(k+1)}$$

- $P(k+1)$ is true.
- By the principle of mathematical induction $P(n)$ is true for all $n \geq 1$.

Number of Subsets of a Finite Set

Example: Use mathematical induction to show that if *S* is a finite set with n elements, where *n* is a nonnegative integer, then *S* has 2^n subsets.

(*Chapter 6 uses combinatorial methods to prove this result.*)

Solution: Let *P*(*n*) be the proposition that a set with *n* elements has 2^n subsets.

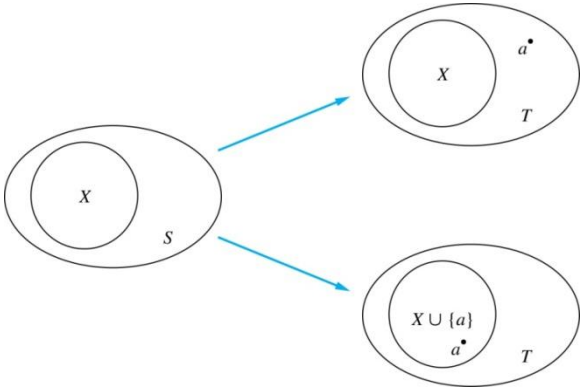
- Basis Step: *P*(0) is true, because the empty set has only itself as a subset and $2^0 = 1$.
- Inductive Step: Assume *P*(*k*) is true for an arbitrary nonnegative integer *k*.

continued →

Number of Subsets of a Finite Set

Inductive Hypothesis: For an arbitrary nonnegative integer k, every set with k elements has 2^k subsets.

- Let T be a set with $k + 1$ elements. Then $T = S \cup \{a\}$, where $a \in T$ and $S = T - \{a\}$. Hence $|T| = k$.
- For each subset X of S, there are exactly two subsets of T, i.e., X and $X \cup \{a\}$.



- By the inductive hypothesis S has 2^k subsets. Since there are two subsets of T for each subset of S, the number of subsets of T is $2 \cdot 2^k = 2^{k+1}$.



Example

- Show by mathematical induction that any finite, nonempty set is countable; that is, it can be arranged in a list.
- Solution
  - Let P(n) be the predicate that if A is any set with $|A|=n\geq 1$, then A is countable.
  - BS: $n_0 = 1$, so we let A be any set with one element, say $A=\{x\}$, In this case x forms a sequence all by itself whose set is A, so P(1) is true.

- IS:
  - assume the statement P(k) that if A is any set with k elements, then A is countable.
  - choose any set B with $k + 1$ elements and pick any element x in B.
  - Since $B - \{x\}$ is a set with k element, the induction hypothesis P (k) tells us there is a sequence $x_1, x_2, \ldots, x_k$ with $B - \{x\}$ as its corresponding set.
  - The sequence $x_1, x_2, \ldots, x_k$, x then has B as the corresponding set. so B is countable.
  - Since B can be any set with k + I elements, P(k + I) is true if P(k) is.
- Thus, by the principle of mathematical induction, P(n) is true for all $n \geq 1$.

Guidelines: Mathematical Induction Proofs

Template for Proofs by Mathematical Induction

1. Express the statement that is to be proved in the form "for all $n \geq b$, $P(n)$" for a fixed integer b.
2. Write out the words "Basis Step." Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words "Inductive Step."
4. State, and clearly identify, the inductive hypothesis, in the form "assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$."
5. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.
6. Prove the statement $P(k + 1)$ making use the assumption $P(k)$. Be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k, including $k = b$.
7. Clearly identify the conclusion of the inductive step, such as by saying "this completes the inductive step."
8. After completing the basis step and the inductive step, state the conclusion, namely that by mathematical induction, $P(n)$ is true for all integers n with $n \geq b$.

Second Principle of Induction

A.k.a. "Strong Induction"

- Characterized by another inference rule:

$P(0)$

$\forall n \geq 0: (\forall 0 \leq k \leq n\ P(k)) \rightarrow P(n+1)$

(*P* is true in *all* previous cases)

$\therefore \forall n \geq 0: P(n)$

- The only difference between this and the 1st principle is that:
  - the inductive step here makes use of the stronger hypothesis that $P(k)$ is true for *all* smaller numbers $k<n+1$, not just for $k=n$.

Strong Induction

- In the strong form of mathematical induction, the preceding（前述的） Inductive Step
  - For every n, *assume* that $S(n)$ is true and *prove* that $S(n+1)$ is true.
- is replaced by
  - For every n, *assume* that $S(k)$ is true *for all* $k < n + 1$ and *prove* that $S(n+1)$ is true.

Example of Second Principle

- Show that every $n>1$ can be written as a product $\prod p_i = p_1 p_2 \ldots p_s$ of some series of s prime numbers.
  - Let $P(n)$="n has that property"
- **Base case:** $n=2$, let $s=1$, $p_1=2$.
- **Inductive step:** Let $n \geq 2$. Assume $\forall 2 \leq k \leq n$: $P(k)$. Consider $n+1$. If it's prime, let $s=1$, $p_1=n+1$. Else $n+1=ab$, where $1<a \leq n$ and $1<b \leq n$. Then $a=p_1 p_2 \ldots p_t$ and $b=q_1 q_2 \ldots q_u$. Then we have that $n+1 = p_1 p_2 \ldots p_t q_1 q_2 \ldots q_u$, a product of $s=t+u$ primes.

Another Example

- Let's prove that every number can be written as the product （乘积）of primes（素数）.
- Clearly, this is true for prime numbers, but how can we prove it in general?

Another Example

- Suppose we have some number n that is the product of two numbers a and b.
- A and B must be less than n, but they can’t be one less... so how can we apply induction?

Products of Primes

- Let P(n) be true if n can be expressed as the product of (one or more) primes.
- Base Case: n = 2
  - 2 is prime, thus P(2) is true

Inductive Step

- Assume $\forall i<n$: P(i) is true
- Case 1: n is prime.
  - P(n) is true. Done!
- Case 2: n is not prime
  - n = ab; 1 < a,b < n
  - By our assumption, P(a) and P(b) are true.
- Therefore n is the product of primes.

Another 2nd Principle Example

- Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps. $P(n)$="n can be…"
- **Base case:** $12=3(4)$, $13=2(4)+1(5)$, $14=1(4)+2(5)$, $15=3(5)$, so $\forall 12\le n\le 15$, $P(n)$.
- **Inductive step:** Let $n\ge 15$, assume $\forall 12\le k\le n\ P(k)$. Note $12\le n-3\le n$, so $P(n-3)$, so add a 4-cent stamp to get postage for $n+1$.

Two ways of looking at Induction

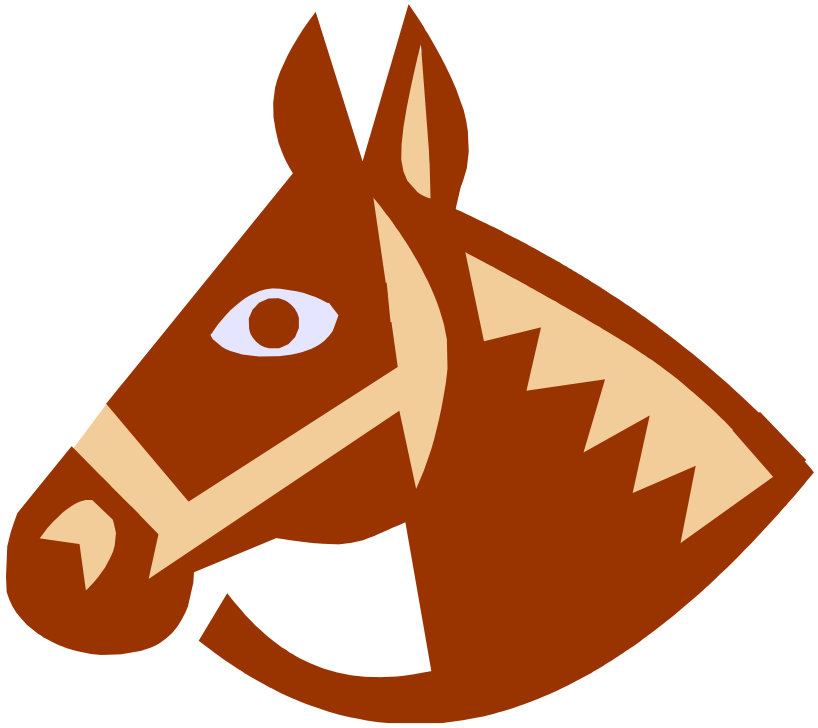
- d_0 hits d_1, d_1 hits d_2 …
- $P(0) \Rightarrow P(1)$, $P(1) \Rightarrow P(2)$ ...
- $P(i) \Rightarrow P(i+1)$

- d_1 falls because d_0 fell,
- d_2 falls since d_0 and d_1 both fell…
- $P(0) \Rightarrow P(1)$, $(P(0) \wedge P(1)) \Rightarrow P(2)$...
- $(\forall i<j\ P(i)) \Rightarrow P(j)$

A Bogus Example

- Theorem: All horses are the same color (!)

- Proof by induction!



An Incorrect "Proof" by Mathematical Induction

Example: Let $P(n)$ be the statement that every set of n lines in the plane, no two of which are parallel, meet in a common point. Here is a "proof" that $P(n)$ is true for all positive integers $n \geq 2$.

- BASIS STEP: The statement $P(2)$ is true because any two lines in the plane that are not parallel meet in a common point.
- INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true for the positive integer $k \geq 2$, i.e., every set of k lines in the plane, no two of which are parallel, meet in a common point.
- We must show that if $P(k)$ holds, then $P(k + 1)$ holds, i.e., if every set of k lines in the plane, no two of which are parallel, $k \geq 2$, meet in a common point, then every set of k + 1 lines in the plane, no two of which are parallel, meet in a common point.

continued →

An Incorrect "Proof" by Mathematical Induction

Inductive Hypothesis: Every set of k lines in the plane, where $k \geq 2$, no two of which are parallel, meet in a common point.

- Consider a set of $k + 1$ distinct lines in the plane, no two parallel. By the inductive hypothesis, the first k of these lines must meet in a common point p_1. By the inductive hypothesis, the last k of these lines meet in a common point p_2.
- If p_1 and p_2 are different points, all lines containing both of them must be the same line since two points determine a line. This contradicts the assumption that the lines are distinct. Hence, $p_1 = p_2$ lies on all $k + 1$ distinct lines, and therefore $P(k + 1)$ holds. Assuming that $k \geq 2$, distinct lines meet in a common point, then every $k + 1$ lines meet in a common point.
- There must be an error in this proof since the conclusion is absurd. But where is the error?
  - **Answer**: $P(k) \rightarrow P(k + 1)$ only holds for $k \geq 3$. It is not the case that $P(2)$ implies $P(3)$. The first two lines must meet in a common point p_1 and the second two must meet in a common point p_2. They do not have to be the same point since only the second line is common to both sets of lines.

Guidelines: Mathematical Induction Proofs

Template for Proofs by Mathematical Induction

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2. Write out the words "Basis Step." Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words "Inductive Step."
4. State, and clearly identify, the inductive hypothesis, in the form "assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$."
5. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.
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7. Clearly identify the conclusion of the inductive step, such as by saying "this completes the inductive step."
8. After completing the basis step and the inductive step, state the conclusion, namely that by mathematical induction, $P(n)$ is true for all integers n with $n \geq b$.

The Well-Ordering Property
良序特性

- Another way to prove the validity of the inductive inference rule is by using the *well-ordering property*, which says that:
  - Every non-empty set of non-negative integers has a minimum (smallest) element.
  - $\forall\, \emptyset \subset S \subseteq \mathbf{N} : \exists m \in S : \forall n \in S : m \leq n$
- This implies that $\{n | \neg P(n)\}$ (if non-empty) has a min. element m, but then the assumption that $P(m-1) \rightarrow P((m-1)+1)$ would be contradicted.

The Method of Infinite Descent
无穷递降法

- A way to prove that $P(n)$ is false for all $n \in \mathbf{N}$.
- Sort of a converse to the principle of induction.
- Prove first that $\forall P(n)$: $\exists k<n$: $P(k)$.
  - Basically, "For every P there is a smaller P."
- But by the well-ordering property of **N**, we know that $\exists P(m) \rightarrow \exists P(n)$: $\forall P(k)$: $n \leq k$.
  - Basically, "If there is a P, there is a smallest P."
- Note that these are contradictory unless $\neg\exists P(m)$,
  - that is, $\forall m \in \mathbf{N}$: $\neg P(m)$. There is no P.

Infinite Descent Example

- **Theorem:** $2^{1/2}$ is irrational.
- **Proof:** Suppose $2^{1/2}$ is rational, then $\exists m,n \in \mathbf{Z}^+$: $2^{1/2}=m/n$. Let M,N be the m,n with the least n.

$$\sqrt{2} = \frac{M}{N} \therefore 2 = \frac{M^2}{N^2} \therefore 2N^2 = M^2.$$

$$\frac{2N-M}{M-N} = \frac{(2N-M)N}{(M-N)N} = \frac{2N^2-MN}{(M-N)N} = \frac{M^2-MN}{(M-N)N} = \frac{(M-N)M}{(M-N)N} = \frac{M}{N}$$

$$1 < \sqrt{2} < 2 \therefore 1 < \frac{M}{N} < 2 \therefore N < M < 2N \therefore 0 < M-N < N$$

So $\exists k<N,j$: $2^{1/2} = j/k$ (let $j=2N-M$, $k=M-N$). ■

作业

- §5.1 – 32, 44, 50, 60
- §5.2 – 4, 26, 32

Recursive Definitions and Structural Induction

Section 5.3

Section Summary

- Recursively Defined Functions
- Recursively Defined Sets and Structures
- Structural Induction
- Generalized Induction

Recursively Defined Functions

Definition: A *recursive* or *inductive definition* of a function consists of two steps.

- BASIS STEP: Specify the value of the function at zero.
- RECURSIVE STEP: Give a rule for finding its value at an integer from its values at smaller integers.

- A function $f(n)$ is the same as a sequence a_0, a_1, … , where a_i, where $f(i) = a_i$. This was done using recurrence relations in Section 2.4.

Recursively Defined Functions

Example: Suppose f is defined by:

$f(0) = 3$,

$f(n + 1) = 2f(n) + 3$

Find $f(1)$, $f(2)$, $f(3)$, $f(4)$

Solution:

- $f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$
- $f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$
- $f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$
- $f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$

Example: Give a recursive definition of the factorial function $n!$:

Solution:

$f(0) = 1$

$f(n + 1) = (n + 1) \cdot f(n)$

Recursively Defined Functions

Example: Give a recursive definition of:

$$\sum_{k=0}^{n} a_k.$$

Solution: The first part of the definition is

$$\sum_{k=0}^{0} a_k = a_0.$$

The second part is $$\sum_{k=0}^{n+1} a_k = \left(\sum_{k=0}^{n} a_k\right) + a_{n+1}.$$

Fibonacci Numbers



Fibonacci
(1170- 1250)

Example : The Fibonacci numbers are defined as follows:

$f_0 = 0$

$f_1 = 1$

$f_n = f_{n-1} + f_{n-2}$

Find f_2, f_3, f_4, f_5.

- $f_2 = f_1 + f_0 = 1 + 0 = 1$
- $f_3 = f_2 + f_1 = 1 + 1 = 2$
- $f_4 = f_3 + f_2 = 2 + 1 = 3$
- $f_5 = f_4 + f_3 = 3 + 2 = 5$

In Chapter 8, we will use the Fibonacci numbers to model population growth of rabbits. This was an application described by Fibonacci himself.

Next, we use strong induction to prove a result about the Fibonacci numbers.


Fibonacci Sequence
1 1 2 3 5 8 13 21 34 55 89 144

Fibonacci Numbers

Example 4: Show that whenever $n \geq 3$, $f_n > \alpha^{n-2}$, where $\alpha = (1+\sqrt{5})/2$.

Solution: Let $P(n)$ be the statement $f_n > \alpha^{n-2}$. Use strong induction to show that $P(n)$ is true whenever $n \geq 3$.

- BASIS STEP: $P(3)$ holds since $\alpha < 2 = f_3$

  $P(4)$ holds since $\alpha^2 = (3+\sqrt{5})/2 < 3 = f_4$.
- INDUCTIVE STEP: Assume that $P(j)$ holds, i.e., $f_j > \alpha^{j-2}$ for all integers j with

  $3 \leq j \leq k$, where $k \geq 4$. Show that $P(k+1)$ holds, i.e., $f_{k+1} > \alpha^{k-1}$.
  - Since $\alpha^2 = \alpha + 1$ (because α is a solution of $x^2 - x - 1 = 0$),

    $\alpha^{k-1} = \alpha^2 \cdot \alpha^{k-3} = (\alpha + 1)\cdot\alpha^{k-3} = \alpha\cdot\alpha^{k-3} + 1\cdot\alpha^{k-3} = \alpha^{k-2} + \alpha^{k-3}$
  - By the inductive hypothesis, because $k \geq 4$ we have

    $f_{k-1} > \alpha^{k-3}$, $f_{k-1} > \alpha^{k-2}$.
  - Therefore, it follows that

    $f_{k+1} = f_{k+1} + f_{k-1} > \alpha^{k-2} + \alpha^{k-3} = \alpha^{k-1}$.
  - Hence, $P(k+1)$ is true.

Why does this equality hold?

Gabriel Lamé
(1795-1870)



Lamé's Theorem拉梅定理

Lamé's Theorem: Let a and b be positive integers with $a \geq b$. Then the number of divisions used by the Euclidian algorithm to find gcd(a,b) is less than or equal to five times the number of decimal digits in b.

Proof: When we use the Euclidian algorithm to find gcd(a,b) with $a \geq b$,

- n divisions are used to obtain (with $a = r_0, b = r_1$):

$$
\begin{aligned}
r_0 &= r_1 q_1 + r_2 && 0 \leq r_2 < r_1,\\
r_1 &= r_2 q_2 + r_3 && 0 \leq r_3 < r_2,\\
&\vdots\\
r_{n-2} &= r_{n-1} q_{n-1} + r_n && 0 \leq r_n < r_{n-1},\\
r_{n-1} &= r_n q_n.
\end{aligned}
$$

- Since each quotient $q_1, q_2, \ldots, q_{n-1}$ is at least 1 and $q_n \geq 2$:

$$
\begin{aligned}
r_n &\geq 1 = f_2,\\
r_{n-1} &\geq 2\, r_n \geq 2\, f_2 = f_3,\\
r_{n-2} &\geq r_{n-1} + r_n \geq f_3 + f_2 = f_4,\\
&\vdots\\
r_2 &\geq r_3 + r_4 \geq f_{n-1} + f_{n-2} = f_n,\\
b = r_1 &\geq r_2 + r_3 \geq f_n + f_{n-1} = f_{n+1}.
\end{aligned}
$$

continued →

Lamé's Theorem

- It follows that if n divisions are used by the Euclidian algorithm to find gcd(a,b) with $a \geq b$, then $b \geq f_{n+1}$. By Example 4, $f_{n+1} > \alpha^{n-1}$, for $n > 2$, where $\alpha = (1+\sqrt{5})/2$. Therefore, $b > \alpha^{n-1}$.
- Because $\log_{10} \alpha \approx 0.208 > 1/5$, $\log_{10} b > (n-1) \log_{10} \alpha > (n-1)/5$. Hence,

$$n-1 < 5 \cdot \log_{10} b.$$

- Suppose that b has k decimal digits. Then $b < 10^k$ and $\log_{10} b < k$. It follows that $n - 1 < 5k$ and since k is an integer, $n \leq 5k$.

- As a consequence of Lamé's Theorem, $O(\log b)$ divisions are used by the Euclidian algorithm to find gcd(a,b) whenever $a > b$.
  - By Lamé's Theorem, the number of divisions needed to find gcd(a,b) with $a > b$ is less than or equal to $5(\log_{10} b + 1)$ since the number of decimal digits in b (which equals $[\log_{10} b] + 1$) is less than or equal to $\log_{10} b + 1$.

Lamé's Theorem was the first result in computational complexity

Recursively Defined Sets and Structures

Recursive definitions of sets have two parts:

- The *basis step* specifies an initial collection of elements.
- The *recursive step* gives the rules for forming new elements in the set from those already known to be in the set.

- Sometimes the recursive definition has an *exclusion rule*, which specifies that the set contains nothing other than those elements specified in the basis step and generated by applications of the rules in the recursive step.
- We will always assume that the exclusion rule holds, even if it is not explicitly mentioned.
- We will later develop a form of induction, called *structural induction*, to prove results about recursively defined sets.

Recursively Defined Sets and Structures

Example : Subset of Integers *S*:

BASIS STEP: $3 \in S$.

RECURSIVE STEP: If $x \in S$ and $y \in S$, then $x + y$ is in *S.*

- Initially 3 is in *S*, then $3 + 3 = 6$, then $3 + 6 = 9$, etc.

Example: The natural numbers **N**.

BASIS STEP: $0 \in \mathbf{N}$.

RECURSIVE STEP: If n is in **N**, then $n + 1$ is in **N**.

- Initially 0 is in *S*, then $0 + 1 = 1$, then $1 + 1 = 2$, etc.

实数是在有理数基础上定义的，有理数又是在整数的基础上定义的，而整数又是在自然数的基础上定义的，那么自然数如何定义呢？

有两个集合 A 和 B，我们称它们为等价的，如果存在一个从 A 到 B 的映射 f，它是1−1的，又是满的。这时我们说 A 和 B 具有相同的势。我们首先承认空集 ϕ 是存在的，考虑一个集合 $\{\phi\}$，它不是空集，凡与 $\{\phi\}$ 等价的集合都有相同的势，我们把 $\{\phi\}$ 简写为1。再考虑集合 $\{\phi,\{\phi\}\}$，它与 $1=\{\phi\}$ 是不等价的，我们把它简写为 2。一般地如果有了 n 之后，可以定义它的跟随 $\{\phi,n\}$，简写为 $n+1$。这样我们就得到了自然数 $\mathbf{N}=\{1,2,3,\cdots,n,\cdots\}$。在 $\mathbf{N}$ 上可以定义加法：$n+m=n+1+1+\cdots+1$，还可以证明加法满足结合律和交换律：$n+(m+p)=(n+m)+p$，$n+m=m+n$。这样我们就从空集出发，定义出自然数 $\mathbf{N}$。这是一个最抽象的定义，比如说 2，它不指二个人，也不指二个物，而是指一个集合 $\{\phi,\{\phi\}\}$，这个集合有两个不同的元素 $\{\phi\}$ 和 ϕ。凡是与它等价的集合，都与它有相同的势，于是二个人，二个物……，都具有相同的势，按我们的理论，用 $\{\phi,\{\phi\}\}$ 作为它们的代表。

Strings

Definition: The set Σ* of *strings* over the alphabet Σ:

BASIS STEP: λ ∈ Σ* (λ is the empty string)

RECURSIVE STEP: If *w* is in Σ* and *x* is in Σ, then *wx* ∈ Σ*.

Example: If Σ = {0,1}, the strings in in Σ* are the set of all bit strings, λ,0,1, 00,01,10, 11, etc.

Example: If Σ = {*a*,*b*}, show that *aab* is in Σ*.

- Since λ ∈ Σ* and *a* ∈ Σ, *a* ∈ Σ*.
- Since *a* ∈ Σ* and *a* ∈ Σ, *aa* ∈ Σ*.
- Since *aa* ∈ Σ* and *b* ∈ Σ, *aab* ∈ Σ*.

String Concatenation

Definition: Two strings can be combined via the operation of *concatenation*. Let Σ be a set of symbols and Σ* be the set of strings formed from the symbols in Σ. We can define the concatenation of two strings, denoted by ·, recursively as follows.

BASIS STEP: If $w \in \Sigma^*$, then $w \cdot \lambda = w$.

RECURSIVE STEP: If $w_1 \in \Sigma^*$ and $w_2 \in \Sigma^*$ and $x \in \Sigma$, then $w_1 \cdot (w_2 x) = (w_1 \cdot w_2)x$.

- Often $w_1 \cdot w_2$ is written as $w_1 w_2$.
- If w_1 = *abra* and w_2 = *cadabra*, the concatenation $w_1 w_2$ = *abracadabra.*

Length of a String

Example: Give a recursive definition of $l(w)$, the length of the string w.

Solution: The length of a string can be recursively defined by:

$l(\lambda) = 0$;

$l(wx) = l(w) + 1$ if $w \in \Sigma^*$ and $x \in \Sigma$.

Balanced Parentheses

Example: Give a recursive definition of the set of balanced parentheses P.

Solution:

BASIS STEP: $() \in P$

RECURSIVE STEP: If $w \in P$, then $() w \in P$, $(w) \in P$ and $w () \in P$.

- Show that (() ()) is in P.
- Why is))(() not in P?

Well-Formed Formulae in Propositional Logic

Definition: The set of *well-formed formulae* in propositional logic involving **T**, **F**, propositional variables, and operators from the set $\{\neg,\wedge,\vee,\rightarrow,\leftrightarrow\}$.

BASIS STEP: **T**,**F**, and *s*, where *s* is a propositional variable, are well-formed formulae.

RECURSIVE STEP: If *E* and *F* are well formed formulae, then $(\neg E)$, $(E \wedge F)$, $(E \vee F)$, $(E \rightarrow F)$, $(E \leftrightarrow F)$, are well-formed formulae.

Examples: $((p \vee q) \rightarrow (q \wedge \mathbf{F}))$ is a well-formed formula.

$pq \wedge$ is not a well formed formula.

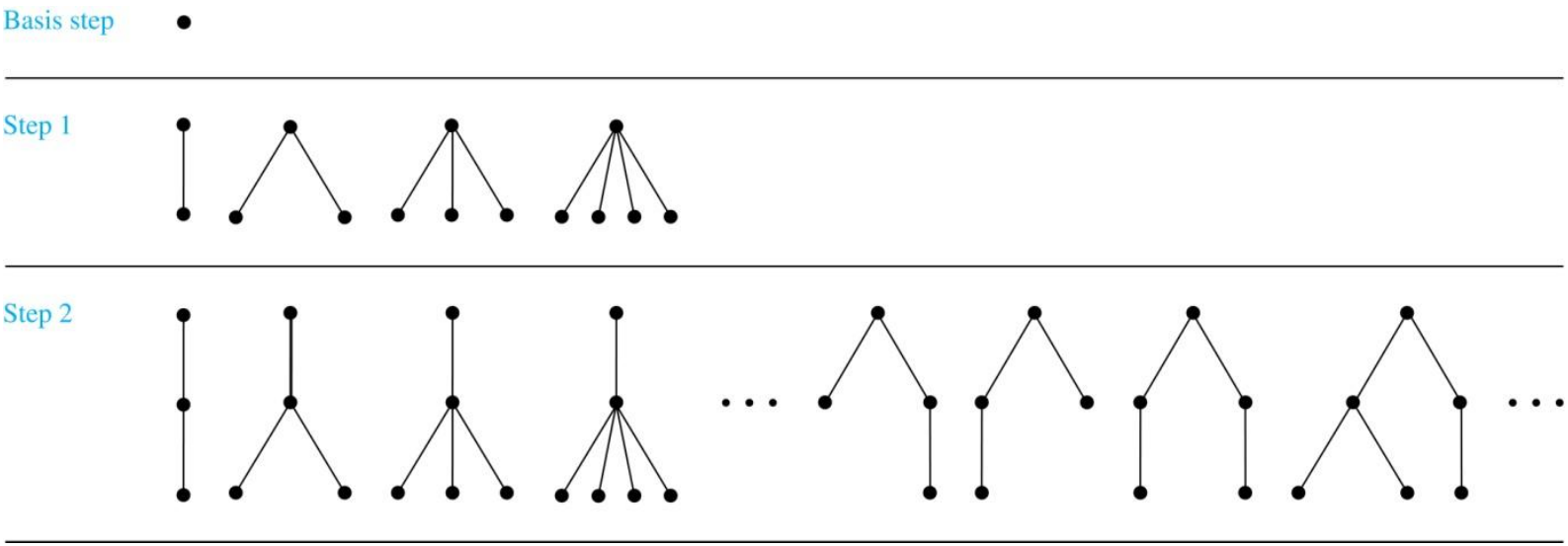
Rooted Trees

Definition: The set of *rooted trees,* where a rooted tree consists of a set of vertices containing a distinguished vertex called the *root*, and edges connecting these vertices, can be defined recursively by these steps:

BASIS STEP: A single vertex r is a rooted tree.

RECURSIVE STEP: Suppose that $T_1, T_2, \ldots, T_n$ are disjoint rooted trees with roots $r_1, r_2, \ldots, r_n$, respectively. Then the graph formed by starting with a root r, which is not in any of the rooted trees $T_1, T_2, \ldots, T_n$, and adding an edge from r to each of the vertices $r_1, r_2, \ldots, r_n$, is also a rooted tree.

Building Up Rooted Trees



- Trees are studied extensively in Chapter 11.
- Next we look at a special type of tree, the full binary tree.

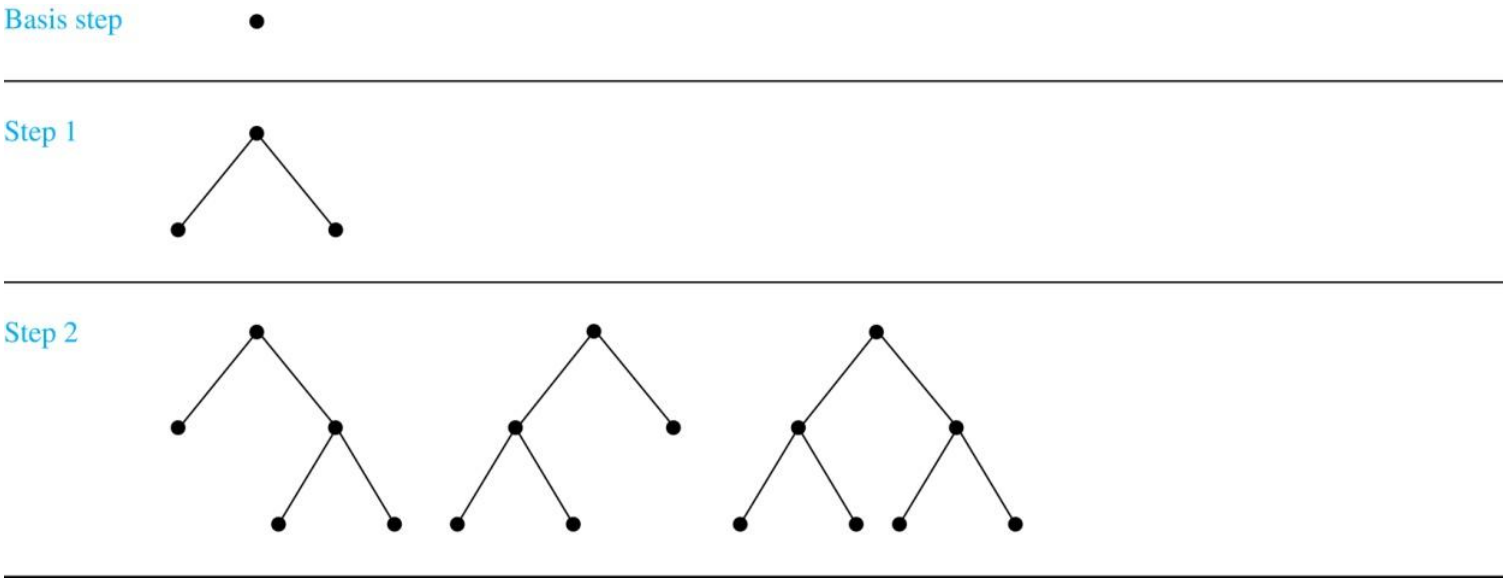
Full Binary Trees

Definition: The set of *full binary trees* can be defined recursively by these steps.

BASIS STEP: There is a full binary tree consisting of only a single vertex *r*.

RECURSIVE STEP: If T_1 and T_2 are disjoint full binary trees, there is a full binary tree, denoted by $T_1 \cdot T_2$, consisting of a root *r* together with edges connecting the root to each of the roots of the left subtree T_1 and the right subtree T_2.

Building Up Full Binary Trees



Induction and Recursively Defined Sets

Example: Show that the set S defined by specifying that $3 \in S$ and that if $x \in S$ and $y \in S$, then $x + y$ is in S, is the set of all positive integers that are multiples of 3.

Solution: Let A be the set of all positive integers divisible by 3. To prove that $A = S$, show that A is a subset of S and S is a subset of A.

- $A \subset S$: Let P(n) be the statement that $3n$ belongs to S.

  BASIS STEP: $3 \cdot 1 = 3 \in S$, by the first part of recursive definition.

  INDUCTIVE STEP: Assume $P(k)$ is true. By the second part of the recursive definition, if $3k \in S$, then since $3 \in S$, $3k + 3 = 3(k+1) \in S$. Hence, $P(k + 1)$ is true.

- $S \subset A$:

  BASIS STEP: $3 \in S$ by the first part of recursive definition, and $3 = 3 \cdot 1$.

  INDUCTIVE STEP: The second part of the recursive definition adds $x + y$ to S, if both x and y are in S. If x and y are both in A, then both x and y are divisible by 3. By part (i) of Theorem 1 of Section 4.1, it follows that $x + y$ is divisible by 3.

- We used mathematical induction to prove a result about a recursively defined set. Next we study a more direct form induction for proving results about recursively defined sets.

Structural Induction

Definition: To prove a property of the elements of a recursively defined set, we use *structural induction*.

BASIS STEP: Show that the result holds for all elements specified in the basis step of the recursive definition.

RECURSIVE STEP: Show that if the statement is true for each of the elements used to construct new elements in the recursive step of the definition, the result holds for these new elements.

- The validity of structural induction can be shown to follow from the principle of mathematical induction.

Full Binary Trees

Definition: The *height h(T)* of a full binary tree *T* is defined recursively as follows:

- BASIS STEP: The height of a full binary tree *T* consisting of only a root *r* is $h(T) = 0$.
- RECURSIVE STEP: If T_1 and T_2 are full binary trees, then the full binary tree $T = T_1 \cdot T_2$ has height $h(T) = 1 + \max(h(T_1), h(T_2))$.

- The number of vertices $n(T)$ of a full binary tree *T* satisfies the following recursive formula:
  - **BASIS STEP**: The number of vertices of a full binary tree *T* consisting of only a root *r* is $n(T) = 1$.
  - **RECURSIVE STEP**: If T_1 and T_2 are full binary trees, then the full binary tree $T = T_1 \cdot T_2$ has the number of vertices $n(T) = 1 + n(T_1) + n(T_2)$.

Structural Induction and Binary Trees

Theorem: If T is a full binary tree, then $n(T) \leq 2^{h(T)+1} - 1$.

Proof: Use structural induction.

- BASIS STEP: The result holds for a full binary tree consisting only of a root, $n(T) = 1$ and $h(T) = 0$. Hence, $n(T) = 1 \leq 2^{0+1} - 1 = 1$.
- RECURSIVE STEP: Assume $n(T_1) \leq 2^{h(T1)+1} - 1$ and also $n(T_2) \leq 2^{h(T2)+1} - 1$ whenever T_1 and T_2 are full binary trees.

$$
\begin{aligned}
n(T) &= 1 + n(T_1) + n(T_2) && \text{(by recursive formula of } n(T)\text{)} \\
&\leq 1 + (2^{h(T1)+1} - 1) + (2^{h(T2)+1} - 1) && \text{(by inductive hypothesis)} \\
&\leq 2 \cdot \max(2^{h(T1)+1}, 2^{h(T2)+1}) - 1 \\
&= 2 \cdot 2^{\max(h(T1),h(T2))+1} - 1 && (\max(2^x, 2^y) = 2^{\max(x,y)}) \\
&= 2 \cdot 2^{h(t)} - 1 && \text{(by recursive definition of } h(T)\text{)} \\
&= 2^{h(t)+1} - 1
\end{aligned}
$$

Generalized Induction

- *Generalized induction* is used to prove results about sets other than the integers that have the well-ordering property. (*explored in more detail in Chapter* 9)
- For example, consider an ordering on **N**× **N**, ordered pairs of nonnegative integers. Specify that (x_1 ,y_1) is less than or equal to (x_2,y_2) if either $x_1 < x_2$, or $x_1 = x_2$ and $y_1 < y_2$. This is called the *lexicographic ordering*.
- Strings are also commonly ordered by a *lexicographic ordering*.
- The next example uses generalized induction to prove a result about ordered pairs from **N**× **N**.

Generalized Induction

Example: Suppose that $a_{m,n}$ is defined for $(m,n) \in \mathbf{N} \times \mathbf{N}$ by $a_{0,0} = 0$ and

$$a_{m,n} = \begin{cases} a_{m-1,n} + 1 & \text{if } n = 0 \text{ and } m > 0 \\ a_{m,n-1} + n & \text{if } n > 0 \end{cases}.$$

Show that $a_{m,n} = m + n(n+1)/2$ is defined for all $(m,n) \in \mathbf{N} \times \mathbf{N}$.

Solution: Use generalized induction.

BASIS STEP: $a_{0,0} = 0 = 0 + (0 \cdot 1)/2$

INDUCTIVE STEP: Assume that $a_{m',n'} = m' + n'(n'+1)/2$ whenever (m',n') is less than (m,n) in the lexicographic ordering of $\mathbf{N} \times \mathbf{N}$.

- If $n = 0$, by the inductive hypothesis we can conclude
  $$a_{m,n} = a_{m-1,n} + 1 = m - 1 + n(n+1)/2 + 1 = m + n(n+1)/2 .$$
- If $n > 0$, by the inductive hypothesis we can conclude
  $$a_{m,n} = a_{m-1,n} + 1 = m + n(n-1)/2 + n = m + n(n+1)/2 .$$

作业

- § 5.1 32, 44, 54
- § 5.2 4, 26
- § 5.3 20, 34,40, 50